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Professional Profile

I am an M.S. candidate in the AI-Embedded Software on Chip Lab at Kyungpook National University, advised by Prof. Daejin Park. I work on **real-time embedded systems and timing-predictable automotive software for Software-Defined Vehicles (SDVs)**. I take control and AI algorithms past the model stage onto automotive MCUs (Infineon AURIX TC375/TC397, NXP S32K344), real motors, in-vehicle networks, and embedded Linux, working down to linker scripts, bootloaders, and device drivers, and measure how they actually behave. My work includes an end-to-end HILS demonstrator for a **Hyundai Motor Company** project, first-author papers at **EMSOFT 2026** (to appear in **IEEE TCAD**) and in **IEEE Access**, first-author presentations at seven international conferences (six oral, one poster), and talks at lab tutorial seminars, the lab symposium, and industry workshops.

Research Interests: Real-Time Embedded Systems · Timing-Predictable Automotive Software · Software-Defined Vehicles · EIL/HIL Validation · Mixed-Criticality Control · HW/SW Co-Design · Embedded AI

EMSOFT'26 / IEEE TCAD | **IEEE Access** | **Hyundai Motor HILS** | **6 Oral + 1 Poster** | **7 SW Copyrights** | **13 Lab Talks**

Education

Kyungpook National University (KNU), Daegu, Korea — Mar 2025 – Feb 2027 (expected)
M.S. in Electronic and Electrical Engineering, School of Electronic and Electrical Engineering — GPA 4.22/4.30
Convergence Major in Future Vehicle Software | Advisor: Prof. Daejin Park | AI-Embedded Software on Chip Lab
Thesis (in progress): *On the Hardware–Software Gap in Runtime Control* (working title)
Coursework: Advanced Operating Systems, Advanced Computer Architecture, Embedded Software Design, Special Topics in Control II, Future Vehicle SW Convergence Seminar I, Industry–Academia Projects

Kyungpook National University (KNU), Daegu, Korea — Mar 2019 – Feb 2025
B.S. in Electronics Engineering, School of Electronics Engineering, College of IT Engineering — GPA 3.72/4.30
Undergraduate researcher, AI-Embedded Software on Chip Lab (2024 – 2025)
Coursework: Operating Systems, Computer Architecture, Microprocessor Design Lab, Data Structures, Automatic Control

Industry–Academia Project

Real-Time HILS for CNN-Based Motor Fault Diagnosis — Mar 2026 – Oct 2026
Hyundai Motor Co. — *PHM Fault-Diagnosis Model Lightweighting & Embedded Methodology*; HILS architecture & integration

- Designed and built the end-to-end HILS: **CARLA vehicle speed drives a real PMSM (NXP S32K344), the teammates' CNN on an AURIX TC397 diagnostic ECU classifies motor signals, and the result feeds back into motor control and CARLA**.
- The model was trained on 25 kHz accelerometer and flux-sensor signals that the rig could not provide, so built vibration/flux proxy signals from motor-control internals in 512 ms windows, sent over a CAN protocol that never disturbs control frames and drops incomplete windows.
- Moved feature extraction onto the TC397 as a streaming Hann + Goertzel stage computing only the 22 required frequencies, and reproduced all five diagnosis classes by injecting fault signatures at the measured-signal stage.
- Added a link-loss fail-safe (0.5 s without CARLA input → self-generated zero-speed command and normal stop; detected in 0.54 s) and a Hyundai-requested fault reaction (three consecutive decisions at $\geq$70% confidence slow or stop the motor and the CARLA vehicle with a warning). Raised the reply rate from 83% to 100% by tracing 8 ms RX-buffer overflows to blocking logging and built-in self-test inference.
- In a 15-minute drive, there were no false decisions without faults, and all 12 injected faults (4 types $\times$ 3) were confirmed. Proxy signals differ from the training data, so results cover the real-time pipeline, not diagnostic accuracy.
- Live-demonstrated it to Hyundai after the final report at the Namyang R&D Center (Oct 16, 2026) on two pre-configured laptops, with a setup manual for Hyundai to rebuild the environment.

Research Experience

On the Hardware–Software Gap in Runtime Control (working title) Aug 2026 – Present
M.S. Thesis — binding-time analysis

- Framing hardware–software partitioning as a choice of *binding time*, the instant a control decision becomes irrevocable, and root-causing fragmented control performance on a Xilinx Zynq-7020 at clock-cycle and register level with paired HW/SW implementations and a bit-exact datapath model.
- Showed that committing a result 160 ns later separated a stable loop from an unstable one, and that the *order* of asynchronous software delays, not only their distribution, decided stability. Designed commit-delay-aware controllers and a judgment-to-use atomic commit.

Command-Authority Scheduling for Mixed-Criticality SDV Control May 2026 – Oct 2026
[J4] IEEE Access (early access, 2026)

- Proposed CAS, which converts low-criticality service outputs into role-constrained authority claims under per-tick ceilings synthesized from an identified plant model and an ISO 15622 / RSS / UN R79 response envelope.
- On a FreeRTOS-based AURIX TC375 ECU-in-the-loop platform with real-time CARLA, CAS was the only service-integrating policy that stayed inside the envelope in all runs. Against a service-free baseline it cut RSS-distance violations by 97% and safety-governor interventions by 88% at a 1.2% lower mean speed, with a worst-case receive-to-transmit time of 171 μs (1.7% of the 10 ms period, debug build).

Reproducible Distributed ECU-in-the-Loop Co-Simulation (JSEC) Oct 2025 – Aug 2026
[J3] EMSOFT 2026 / IEEE TCAD

- Built a distributed CARLA–Simulink (14-DOF vehicle dynamics)–AURIX TC375 testbed over UDP, with a byte-level TRIG/ACT/LOG packet protocol and non-blocking logging.
- Proposed JSEC, which maps admitted ECU events to fixed-step plant advancement with bounded catch-up execution. Raised mean R^2 across repeated runs from 0.410 to 0.947 with 0.79–1.14% execution-time overhead.

On-Chip ADC Variability for Early Detection of Software-Induced Margin Degradation Dec 2025 – Jun 2026
[C6] IPFA 2026

- Designed five software-induced stress scenarios on an AURIX TC375 and showed that stress first appears as increased ADC output *variability* while ISR runtime and ADC means stay nominal.
- Defined a 3σ threshold that separates latent stress from environmental noise, with no extra sensors or hardware changes.

Architecture-Aware Neural Firmware Compression via Knowledge Distillation Jun 2025 – Jan 2026
[C5] ICAIIC 2026

- Distilled a Transformer teacher into a lightweight GRU student that predicts the next byte of TriCore firmware and drives an arithmetic coder, trained on about 60 TC375 firmware images.
- Compressed firmware to 19.05% of its original size, about 38% better than LZMA, with a 0.25 MB decoding footprint (about 94% lower). Python-vs-C cross-validation confirmed bit-exact decoding.

Earlier Research: Embedded Vehicle Control, Perception, and Emulation Jan 2024 – Sep 2025
[C1]–[C4], [J1], [J2]

- **Adaptive sliding-mode control** [C4]: tuned Super-Twisting gains online with a rule-based empirical RL update (no neural-network training) and blended linear and nonlinear sliding surfaces.
- **Multithreaded PID control** [C3]: parallelized sensing, computation, and actuation with precomputed control candidates in portable C on QEMU–Simulink, cutting average settling steps by about 32.3%.
- **Camera-only depth mapping** [C2, J2]: LiDAR-free stereo depth maps with a custom 3×3 convolution kernel and shadow removal, at real-time rates without deep-learning models.
- **ADAS emulation and adaptive parking** [C1, J1]: Smart Cruise Control and Lane Keeping Assist with YOLO detection and ESP32 control, plus a driver-adaptive four-camera parking assistant.

Engineering Projects

TC375 Bootloader and Linker-Script Partitioning Aug 2025
TASKING linker script (LSL), startup code, boot flow

- Analyzed the TC375 TASKING linker script and split it into a bootloader image and an application image; prototyped a minimal bootloader that checks the application entry, disables interrupts, and jumps to the application.

TriCore Firmware Binary Analysis and Runtime Code Loading Aug 2025
Linker maps, Intel HEX, instruction encoding

- Patched variable initial values and retargeted a function call directly in Intel HEX without rebuilding, and verified the change on the board.
- XOR-encrypted a function body in Flash, then decrypted and copied it into RAM at runtime and executed it through a function pointer (a TRAP occurs without decryption).

Simulink MBD-to-AURIX Code Integration Automation Jul 2025

Model-based design, code generation, Python tooling

- Wrote Python tools that parse Embedded Coder output by function-name and comment patterns, insert TC375 PWM/ADC driver code, and generate the main source file, replacing manual integration.
- Validated Sine→PWM and ADC→PID→PWM models end to end on a real AURIX TC375 board.

Linux UART Device Driver with CRC/ECC and Huffman Coding Dec 2024 – Jan 2025

Lab coding camp — team lead (5 members), integration & debugging

- Wrote a GPIO-based UART driver for two BeagleBone boards (Debian, kernel 4.19) as a loadable kernel module exposing a character device: a falling-edge IRQ starts an `hrtimer` that samples each bit and re-arms with `hrtimer_forward_now()`, with the sender and receiver timers aligned to the same bit time.
- Appended SEC-DED (7 bit) and CRC16 to each 32-bit block; the receiver runs SEC-DED only when CRC fails, and a receive log showed a CRC-failed block recovered. With Huffman coding, 1-bit errors were detected but not corrected (variable-length bits broke the fixed-block checks); identified per-block CRC on the compressed stream as the remedy (not implemented) and built a double-buffered encode/transmit pipeline (two threads, mutex, condition variable), leaving UART integration and performance comparison as future work.

uClinux Bring-Up on STM32F429 Nov 2024

Embedded Linux, bootloader, toolchain

- Built and flashed U-Boot, an XIP kernel, and ROMFS on an STM32F429I-DISC1, fixing host toolchain issues, ran cross-compiled BFLT applications from a UART console, and traced the VMA/LMA layout and the BFLT loader's Flash-to-RAM relocation on a no-MMU system.

Adaptive Auto-Parking Control: PID, MPC, and Disturbance Compensation Aug 2024 – Sep 2024

Control design in MATLAB/Simulink and C

- Implemented gain-scaled adaptive PID (Simulink) and MPC (MATLAB and C), and added Kalman-filtered disturbance compensation that suppresses control spikes at disturbance onset.

Teaching & Technical Seminars

Guest Lecturer & Lab Assistant, LIG Nex1 Embedded Systems Workshop Sep 2025

Industry workshop organized by Prof. Daejin Park

- Delivered the uClinux session (building uClinux on STM32F429, explaining XIP execution and BFLT relocation with `objdump`) and ran the TRACE32 hands-on demo on a TC375 (breakpoints and peripheral views).

Teaching Assistant, Embedded Software Lectures for Industry Engineers Multiple sessions

Hyundai Motor Company, Hyundai AutoEver, LIG Nex1 (basic/advanced) — lab setup and hands-on support

Lab Tutorial Seminars & Symposium Talks Jul 2024 – Sep 2025

AI-Embedded Software on Chip Lab — 13 hands-on presentations with board results and troubleshooting notes

- **MCU firmware & debugging:** TRACE32 (ARM & TriCore), Infineon MemTool, TriCore HEX modification, Simulink MBD on TriCore, NXP S32K344 PMSM FOC, STM32–Simulink.
- **Embedded Linux & virtualization:** uClinux on STM32F429 (U-Boot, XIP kernel, BFLT loader), QEMU/KVM setup, Simulink–QEMU (Cortex-M3 ELF, startup code, linker script).
- **Control & simulation:** MPC in MATLAB and C, linking MATLAB & C (MEX, Engine API, Coder), reinforcement learning in MATLAB, CARLA simulator.

Publications

IEEE TCAD (EMSOFT) ×1 **IEEE Access ×1** **KCI Journal ×2** **Int'l Conference ×6** First author on all 10

Journal Articles

[J4] **H. Lee** and D. Park, "Command-Authority Scheduling for Bounded Low-Criticality Service Influence in Mixed-Criticality SDV Control," *IEEE Access*, early access, 2026, doi: 10.1109/ACCESS.2026.3749257.

[J3] **H. Lee** and D. Park, "JSEC: Jitter-to-Step Execution Contracts for Reproducible Distributed ECU-in-the-Loop Co-Simulation," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, to appear, doi: 10.1109/TCAD.2026.3722824.
EMSOFT 2026 full paper, journal track (ACM/IEEE Int. Conf. on Embedded Software), accepted Aug. 2026; presented at ESWEEK 2026, Barcelona, Spain, Oct. 7, 2026. (Oral)

[J2] **H. Lee** and D. Park, "Real-Time Depth Map Using Stereo Vision and Convolutional Filtering," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 29, no. 9, Sep. 2025. (KCI)

[J1] **H. Lee** and D. Park, "Auto Parking Assistance Control using Human-Activity Feedback-based Embedded Software Emulation," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 28, 2024. (KCI)

International Conference Papers

[C6] **H. Lee** and D. Park, "Early Detection of Software-Induced Internal Margin Degradation in Automotive MCUs Using On-Chip ADC Variability," in *Proc. IEEE Int. Symp. on the Physical and Failure Analysis of Integrated Circuits (IPFA)*, Singapore, Jul. 2026. (Oral)

[C5] **H. Lee**, H. Kim, and D. Park, "Architecture-Aware Neural Compression for TriCore Firmware using Knowledge Distillation," in *Proc. Int. Conf. on Artificial Intelligence in Information and Communication (ICAIIC)*, Tokyo, Japan, Feb. 2026. (Oral)

[C4] **H. Lee** and D. Park, "Lightweight Empirical Reinforcement Learning Driven Adaptive Super-Twisting Control with Fused Linear-Nonlinear Sliding Surfaces for Embedded Vehicle Control," in *Proc. 51st Annual Conf. of the IEEE Industrial Electronics Society (IECON)*, Madrid, Spain, Oct. 2025. (Oral)

[C3] **H. Lee** and D. Park, "Advanced Real-Time Performance QEMU Emulated Automotive Control System with a Multithreaded PID Controller," in *Proc. IEEE Int. Conf. on Consumer Electronics – Taiwan (ICCE-TW)*, Kaohsiung, Taiwan, Jul. 2025. (Oral)

[C2] **H. Lee** and D. Park, "Convolution-Based Depth Map With Shadow Removal Using Cameras for 3D Mapping in Autonomous Vehicle Driving," in *Proc. IEEE Int. Symp. on Intelligent Signal Processing and Communication Systems (ISPACS)*, Kaohsiung, Taiwan, Dec. 2024. (Oral)

[C1] **H. Lee** and D. Park, "Multi-Camera Interoperable Emulation Framework Using Embedded Edge-Cloud AI Computing for Autonomous Vehicle Driving," in *Proc. IEEE Vehicular Networking Conf. (VNC)*, Kobe, Japan, May 2024. (Poster)

Software Copyright Registrations

Korea Copyright Commission; holder: KNU Industry–Academic Cooperation Foundation; co-developed with Prof. D. Park.

[S7]	TriCore-Based Intel HEX Firmware Compression, No. C-2025-044871	Oct 2025 · [C5]
[S6]	Real-Time Depth Map Using Stereo Vision, No. C-2025-044868	Oct 2025 · [J2]
[S5]	Adaptive Super-Twisting Control, No. C-2025-044867	Oct 2025 · [C4]
[S4]	QEMU-Based Multi-Threaded Automotive Control, No. C-2025-044866	Oct 2025 · [C3]
[S3]	Depth Map Generation Program for 3D Mapping, No. C-2024-054193	Dec 2024 · [C2]
[S2]	Autonomous Driving Emulation Program, No. C-2024-052879	Dec 2024 · [C1]
[S1]	Parking Assistance Control Adjustment, No. C-2024-051914	Dec 2024 · [J1]

Technical Skills

Programming	C, C++, Python, MATLAB, Bash; reading ARM and TriCore assembly; Makefiles and linker scripts
MCUs & Boards	AURIX TC375/TC397 (TriCore), NXP S32K344, STM32F429, Xilinx Zynq-7020, BeagleBone
Firmware & OS	Bootloaders, linker scripts (TASKING LSL, GNU ld), startup code, Timers/ISR, PWM, ADC, DMA, FreeRTOS, uClinux, Linux kernel-module device drivers, cross-compilation, ELF/map/Intel HEX
Networks	CAN, UART, UDP, TCP/IP, Ethernet (LwIP)
Debug & Flash	TRACE32, Infineon MemTool, FreeMASTER, AURIX Dev. Studio, S32DS, EB tresos, OpenOCD
Simulation	MATLAB/Simulink, Embedded Coder, CARLA, QEMU/KVM, ROS/ROS 2 (CARLA bridge setup)
Control	PID, MPC, sliding-mode/super-twisting control, PMSM FOC, Kalman filter, disturbance observer
AI/ML & Vision	PyTorch, RL (DQN, PPO), imitation learning, knowledge distillation, OpenCV, YOLO, Open3D

Languages, Certifications & Other

- **Languages:** Korean (native), English (OPIc Intermediate High, ACTFL) — May 2026
- **Computer Specialist in Spreadsheet & Database, Level 1**, Korean national certification (KCCI) — Oct 2020
- **XR Expert Training Program** (48 h, Unity/C#), Daegu VR·AR Production Center — Oct 2022 – Dec 2022
- **Military Service:** completed as Social Service Personnel — Jan 2021 – Oct 2022

References

Prof. Daejin Park (M.S. Advisor), School of Electronics Engineering, KNU — boltanut@knu.ac.kr